

Oracle RDS Data guard

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Overview

PSP uses 2 databases (Monolith & Audit/IBOB). Monolith has read replica in same region(us-west-2), which is used by downstream services to pull data and by RJF application to generate reports. But this cannot be used as DR because till now it did not have switchover/switchback capability. Hence, there is need to maintain separate databases for OLAP use-cases and DR (GoldenGate Replication).

Problem Statement

- High DR replication lag(~10 hrs during peak day) and hence High RTO for DR scenario.
- Need to maintain separate read replica database for OLAP use cases.
- High RDS infrastructure Cost

Goal

- Build DR using Data Guard technology with different configuration for Primary and DR
- Switchover & Switchback capability

Benefits

- Cost reduction for RDS infrastructure
- Reduce replication lag to near zero(in milliseconds).
- Best option to serve for DR as well as OLAP use cases.

Implementation

Database Version **19c** was used for below.

Actions	Primary Configuration	Standby Configuration	Actions	Status	Elapsed Time
Create	instance size: db.r5.large iops: 1000 storage: 100G	instance size: db.r5.2xlarge iops: 1000 storage: 100G	Switchover & Switchback	Successful	~5 mins
Config Change	instance size: db.r5.large iops: 1000 storage: 100G	instance size: db.r5.2xlarge iops: 3000 storage: 200G	Switchover & Switchback	Successful	~5 mins
Config Change	instance size: db.r5.large iops: 1000 storage: 100G	1. instance size: db.r5.2xlarge iops: 3000 storage: 200G 2. instance size: db.r5.large iops: 1000 storage: 100G	Switchover to 1st standby Switchover to 2nd standby Switchback to Original Primary	Successful	

Observations/Limitations/Restrictions

- Backup needs to be enabled and retention should be there at least 1 day. Otherwise creation of cross-Region replica/role reversal will fail.

```
mchoubey@:intuitdep626ed:~/intuit/sbg/psp/aws/dataguard->aws --profile sbg-qbo-ppd --region us-east-2 rds
switchover-read-replica --db-instance-identifier eqbodg69
```

An error occurred (InvalidDBInstanceState) when calling the SwitchoverReadReplica operation: The new primary DB instance isn't ready for a role transition. The instance must be available, have no operations pending in its maintenance window, and have a backup retention period greater than 0.

- Option group on Primary should not be part of any other Data Guard configuration/Source having Read Replica. Option group can be part of any other individual instances.

```
mchoubey@:intuitdep626ed:~/intuit/sbg/psp/aws/dataguard->aws --profile sbg-qbo-ppd --region us-east-2 rds
switchover-read-replica --db-instance-identifier eqbodg69
```

An error occurred (InvalidDBInstanceState) when calling the SwitchoverReadReplica operation: You can't switch over because the primary instance's (eqbott69) option group contains unrelated instances.

- When you create an Oracle cross-Region replica, it does not migrate existing option group of source rds instance, it creates a dedicated option group for it. You can't remove an Oracle cross-Region replica from its dedicated option group. No other DB instances can use the dedicated option group for an Oracle cross-Region replica. You can only add or remove options from a dedicated option group (for eg: `NATIVE_NETWORK_ENCRYPTION`, `Timezone` etc.)
- **There is no FailOver option available at this moment as FLASHBACK is not enabled. This would be enabled by Q4 2022.**
- **Cannot create RESTORE POINT on standby as FLASHBACK is not enabled.**

Switchover

```
aws --profile sbg-qbo-ppd --region us-east-2 rds switchover-read-replica --db-instance-identifier eqbodg69
{
  "DBInstance": {
    "DBInstanceIdentifier": "eqbodg69",
    "DBInstanceClass": "db.r5.2xlarge",
    "Engine": "oracle-ee",
    "DBInstanceStatus": "available",
    "MasterUsername": "intuadmin",
    "DBName": "EQBOTT69",
    "Endpoint": {
      "Address": "eqbodg69.cdcigllehyqw.us-east-2.rds.amazonaws.com",
      "Port": 1521,
      "HostedZoneId": "Z2XHWR1WZ565X2"
    },
    "AllocatedStorage": 100,
    "InstanceCreateTime": "2022-03-24T08:17:10.691000+00:00",
    "PreferredBackupWindow": "09:57-10:27",
    "BackupRetentionPeriod": 1,
    "DBSecurityGroups": [],
    "VpcSecurityGroups": [
      {
        "VpcSecurityGroupId": "sg-d1164eba",
        "Status": "active"
      }
    ],
    "DBParameterGroups": [
```

```

    {
      "DBParameterGroupName": "oracle-ee-19-max-string-size-e2e-clusterdb-vpc-2",
      "ParameterApplyStatus": "in-sync"
    }
  ],
  "AvailabilityZone": "us-east-2b",
  "DBSubnetGroup": {
    "DBSubnetGroupName": "e2e-clusterdb-vpc-2",
    "DBSubnetGroupDescription": "e2e-clusterdb-vpc-2 subnet group",
    "VpcId": "vpc-1a6c0c73",
    "SubnetGroupStatus": "Complete",
    "Subnets": [
      {
        "SubnetIdentifier": "subnet-f9304690",
        "SubnetAvailabilityZone": {
          "Name": "us-east-2a"
        },
        "SubnetOutpost": {},
        "SubnetStatus": "Active"
      },
      {
        "SubnetIdentifier": "subnet-caf33f87",
        "SubnetAvailabilityZone": {
          "Name": "us-east-2c"
        },
        "SubnetOutpost": {},
        "SubnetStatus": "Active"
      },
      {
        "SubnetIdentifier": "subnet-a971cdd2",
        "SubnetAvailabilityZone": {
          "Name": "us-east-2b"
        },
        "SubnetOutpost": {},
        "SubnetStatus": "Active"
      }
    ]
  },
  "PreferredMaintenanceWindow": "fri:09:27-fri:09:57",
  "PendingModifiedValues": {},
  "MultiAZ": false,
  "EngineVersion": "19.0.0.0.ru-2021-10.rur-2021-10.r1",
  "AutoMinorVersionUpgrade": false,
  "ReadReplicaSourceDBInstanceIdentifier": "arn:aws:rds:us-west-2:475968958217:db:eqbott69",
  "ReadReplicaDBInstanceIdentifiers": [],
  "ReplicaMode": "open-read-only",
  "LicenseModel": "bring-your-own-license",
  "Iops": 1000,
  "OptionGroupMemberships": [
    {
      "OptionGroupName": "replica-19-f92f8d35-4b68-4443-bc9e-134dd50de921",
      "Status": "in-sync"
    }
  ],
  "CharacterSetName": "AL32UTF8",
  "NcharCharacterSetName": "AL16UTF16",
  "PubliclyAccessible": false,
  "StatusInfos": [
    {
      "StatusType": "read replication",
      "Normal": true,
      "Status": "replicating"
    }
  ],
  "StorageType": "io1",
  "DbInstancePort": 0,
  "StorageEncrypted": true,
  "KmsKeyId": "arn:aws:kms:us-east-2:475968958217:key/558d2a3e-d325-4fba-8f7f-4fd9e1fd5913",
  "DbiResourceId": "db-BK7446P270DAZN74K372BJSOXU",
  "CACertificateIdentifier": "rds-ca-2019",
  "DomainMemberships": [],

```

```

    "CopyTagsToSnapshot": false,
    "MonitoringInterval": 0,
    "DBInstanceArn": "arn:aws:rds:us-east-2:475968958217:db:eqbodg69",
    "IAMDatabaseAuthenticationEnabled": false,
    "PerformanceInsightsEnabled": false,
    "EnabledCloudwatchLogsExports": [
        "alert"
    ],
    "DeletionProtection": false,
    "AssociatedRoles": [],
    "TagList": [],
    "CustomerOwnedIpEnabled": false,
    "BackupTarget": "region"
}
}

```

Switchback

```

aws --profile sbg-qbo-ppd --region us-west-2 rds switchover-read-replica --db-instance-identifier eqbott69
{
  "DBInstance": {
    "DBInstanceIdentifier": "eqbott69",
    "DBInstanceClass": "db.r5.large",
    "Engine": "oracle-ee",
    "DBInstanceStatus": "available",
    "MasterUsername": "intuadmin",
    "DBName": "EQBOTT69",
    "Endpoint": {
      "Address": "eqbott69.cozwkqglitfx.us-west-2.rds.amazonaws.com",
      "Port": 1521,
      "HostedZoneId": "Z1PVI0B656C1W"
    },
    "AllocatedStorage": 100,
    "InstanceCreateTime": "2022-03-24T06:50:43.924000+00:00",
    "PreferredBackupWindow": "09:57-10:27",
    "BackupRetentionPeriod": 1,
    "DBSecurityGroups": [],
    "VpcSecurityGroups": [
      {
        "VpcSecurityGroupId": "sg-c2fd32b8",
        "Status": "active"
      }
    ],
    "DBParameterGroups": [
      {
        "DBParameterGroupName": "oracle-ee-19-max-string-size-e2e-clusterdb-vpc-3",
        "ParameterApplyStatus": "in-sync"
      }
    ],
    "AvailabilityZone": "us-west-2b",
    "DBSubnetGroup": {
      "DBSubnetGroupName": "e2e-clusterdb-vpc-3",
      "DBSubnetGroupDescription": "e2e-clusterdb-vpc-3 subnet group",
      "VpcId": "vpc-83f115e5",
      "SubnetGroupStatus": "Complete",
      "Subnets": [
        {
          "SubnetIdentifier": "subnet-04d7045f",
          "SubnetAvailabilityZone": {
            "Name": "us-west-2c"
          },
          "SubnetOutpost": {},
          "SubnetStatus": "Active"
        },
        {
          "SubnetIdentifier": "subnet-66521a2f",
          "SubnetAvailabilityZone": {

```

```

        "Name": "us-west-2b"
      },
      "SubnetOutpost": {},
      "SubnetStatus": "Active"
    },
    {
      "SubnetIdentifier": "subnet-2fc1f048",
      "SubnetAvailabilityZone": {
        "Name": "us-west-2a"
      },
      "SubnetOutpost": {},
      "SubnetStatus": "Active"
    }
  ]
},
"PreferredMaintenanceWindow": "fri:09:27-fri:09:57",
"PendingModifiedValues": {},
"MultiAZ": false,
"EngineVersion": "19.0.0.0.ru-2021-10.rur-2021-10.r1",
"AutoMinorVersionUpgrade": false,
"ReadReplicaSourceDBInstanceIdentifier": "arn:aws:rds:us-east-2:475968958217:db:eqbodg69",
"ReadReplicaDBInstanceIdentifiers": [],
"ReplicaMode": "open-read-only",
"LicenseModel": "bring-your-own-license",
"Iops": 1000,
"OptionGroupMemberships": [
  {
    "OptionGroupName": "tt69-oracle-ee-19-e2e-clusterdb-vpc-3",
    "Status": "in-sync"
  }
],
"CharacterSetName": "AL32UTF8",
"NcharCharacterSetName": "AL16UTF16",
"PubliclyAccessible": false,
"StatusInfos": [
  {
    "StatusType": "read replication",
    "Normal": true,
    "Status": "replicating"
  }
],
"StorageType": "io1",
"DbInstancePort": 0,
"StorageEncrypted": true,
"KmsKeyId": "arn:aws:kms:us-west-2:475968958217:key/7bfb025e-4bf2-4be0-acd8-e58136b2df38",
"DbiResourceId": "db-YYBNHZN3DPAB77SDZ0R407LD44",
"CACertificateIdentifier": "rds-ca-2019",
"DomainMemberships": [],
"CopyTagsToSnapshot": true,
"MonitoringInterval": 0,
"DBInstanceArn": "arn:aws:rds:us-west-2:475968958217:db:eqbott69",
"IAMDatabaseAuthenticationEnabled": false,
"PerformanceInsightsEnabled": true,
"PerformanceInsightsKMSKeyId": "arn:aws:kms:us-west-2:475968958217:key/7bfb025e-4bf2-4be0-acd8-e58136b2df38",
"PerformanceInsightsRetentionPeriod": 7,
"DeletionProtection": true,
"AssociatedRoles": [],
"TagList": [],
"CustomerOwnedIpEnabled": false,
"BackupTarget": "region"
}
}

```

- Database status as per switchover and switchback

DB_UNIQUE_NAME	OPEN_MODE	LOG_MODE	DATABASE_ROLE	FLASHBACK_ON
EQB0TT69_A	READ WRITE		PRIMARY	

INSTANCE	OPEN_MODE	LOG_MODE	DATABASE_ROLE	FLASHBACK_ON
HOST_NAME		STATUS	STARTUP_TIME	

EQB0TT69	READ WRITE	ARCHIVELOG	PRIMARY	NO	ip-172-19-2-
74		OPEN	03/24/2022 12:10:05		

DB_UNIQUE_NAME	OPEN_MODE	LOG_MODE	DATABASE_ROLE	FLASHBACK_ON
EQB0TT69_B	READ ONLY WITH APPLY PHYSICAL STANDBY			

INSTANCE	OPEN_MODE	LOG_MODE	DATABASE_ROLE	FLASHBACK_ON
HOST_NAME		STATUS	STARTUP_TIME	

EQB0TT69	READ ONLY WITH APPLY	ARCHIVELOG	PHYSICAL STANDBY	NO	ip-172-31-3-
154		OPEN	03/24/2022 12:19:41		